

SP250 Homework 1

1. Let a and $b \neq 0$ be integers, such that a is divisible by b . Prove rigorously that for any integer c , $a \cdot c$ is divisible by b (In other words, $b \mid a \cdot c$).

Solution:

By definition, because a is divisible by b , that means that there exists an integer d such that $a = b \cdot d$. But then:

$$a \cdot c = (b \cdot d) \cdot c = b \cdot (d \cdot c)$$

$d \cdot c$ is an integer because both d and c are, thus $b \mid (a \cdot c)$ by definition.

2. What is the sum of all the odd positive integers less than 1000?

Solution:

There are 500 odd positive integers less than 1000. If x is odd then so is $(1000 - x)$. Hence these 500 odd integers can be paired as follows: (1, 999), (3, 997), (5, 995), ... , such that the sum in each pair adds up to 1000.

Therefore, the total sum is equal to $1000 \cdot 500/2 = 250,000$

3. The country of Mathlandia has 10 major cities, any two of which are connected by exactly one road. How many roads are there in total between those 10 cities?

Solution 1 (slick): For every road, put an arrow next to the two cities that it is connecting. Since every two cities are connected by a road, there are 9 arrows next to every city and thus there are a total of $9 \cdot 10 = 90$ arrows. Because there are two arrows for every road, there are a total of $90/2 = 45$ roads.

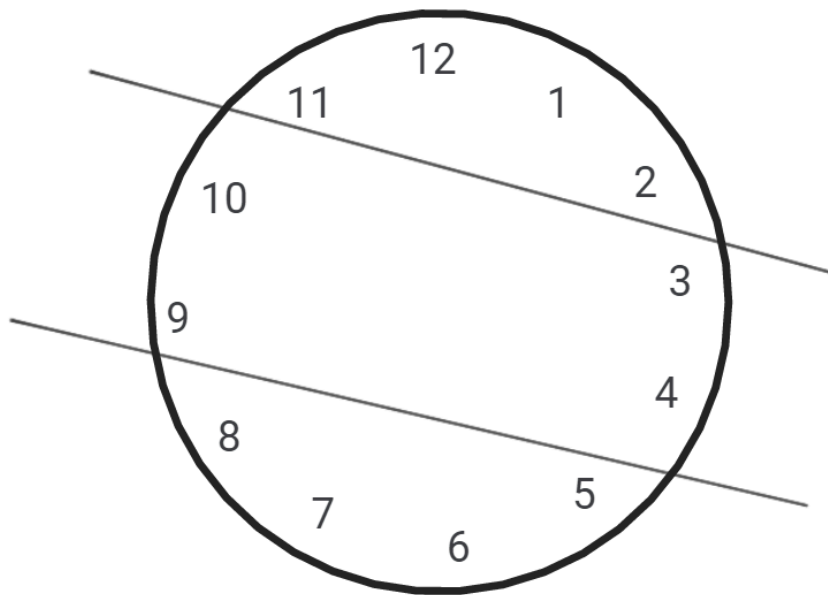
Solution 2: Pick a city. It has 9 roads coming out of it. Erase them. Pick the next city; it now has 8 roads coming out of it. Etc. Hence the total number of roads is

$$9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 + 0 = \frac{9 \cdot 10}{2} = 45$$

4. Numbers 1, 2, ... 11, 12 are written in order along the edge of a round disk (like a watch). Cut the disk into 3 pieces (by straight lines) so that the sum of numbers in each piece is the same.

Solution:

Using the rainbow method (or by simply adding them all), the sum of the numbers around the clock is $(1+12) \cdot 12 / 2 = 78$ and so each of the pieces has to sum up to $78 / 3 = 26$. One way to get that sum is $11+12+1+2$, but then the next block of numbers won't sum up to 12. To get around that problem, we notice that one piece can have two blocks of numbers and we can do it as shown in the figure below.



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5. All integers between 1 and 999 were written out into one very large number. How many 0's were used? Explain your reasoning.

Solution 1:

Before 100, there are exactly 9 0's used – 10, 20, 30, ..., 90.

Between 100 and 199, there are exactly 20 0's used – 10 as the units digit (100, 110, 120, ..., 190) and 10 as the tens digit (100, 101, 102, ..., 109).

Between 200 and 299 it's also 20 0's, etc., up to 900 – 999.

Therefore, the answer is $9 + 20 * 9 = 189$.

Solution 2:

The numbers 1, 2, 3, ..., 999 contain 9 one-digit numbers, 90 two-digit numbers, and 900 three-digit numbers, for a total of 2889 digits. Alternatively, if we write 000, 001, 002, 003, ..., 999, there are a total of 3000 digits, with every digit (including zero) appearing 300 times. The difference of $3000 - 2889 = 111$ digits between the two sets of numbers is all due to extra zeros in the second set. Hence, the digit zero appears 111 fewer times in the first set of numbers than in the second set. So there are $300 - 111 = 189$ zeros in the first set of numbers.

6. Two piles of stones are given. One pile has 10 stones while the other pile has 33 stones. Two players take turns playing the game where, in turn, each player may take 1, 2, 3 or 4 stones from one of the piles. The player who takes the last stone wins. Which player has a winning strategy, first or second?

Solution:

It is the first player that has the winning strategy. He takes 3 stones from the pile with 33 stones (he can also win by taking 2 stones from the other pile but the strategy is a little harder to trace in that case – it is keeping the number of stones so that they have the same remainder when divided by 5). After he does so, both piles have a multiple of 5 stones in them and the first player (who is now second) will use the 1-pile strategy of complementing the moves of the other player to 5 stones.
